

AI-Based Disease Detection System

Structured technical document aligned for formal review, sign-off, and handover.

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Revision History

Name	Date	Reason For Changes	Version
Owais	2026-04-16	Initial generated draft for review.	1.0

1: Introduction

1.1: Purpose

Purpose: To develop an AI-based system for detecting plant diseases from leaf images uploaded by farmers.

1.2: Scope

Scope: This document specifies the requirements for the AI-Based Disease Detection System. It covers functionalities related to image analysis, treatment suggestions, and historical tracking but excludes aspects such as security, scalability, and maintainability.

1.3: Definitions/Acronyms

- **AI:** Artificial Intelligence
- **ML:** Machine Learning

1.4: References

References: IEEE Std 830-1998, IEEE Standard for Software Requirements Specifications

1.5: Overview

Overview: The system is designed to assist farmers in managing their crops more effectively by providing timely information about potential diseases.

2: Overall Description

2.1: Product Perspective

Product Perspective: The product is intended for agricultural users who need quick and accurate identification of plant diseases. The target audience includes small-scale and large-scale farmers.

2.2: Product Functions

- **Image Upload:** Allow farmers to upload leaf images via a mobile or web application.
- **Image Analysis:** Use ML models to detect diseases present in the uploaded images.
- **Treatment Suggestions:** Provide recommendations based on identified diseases.
- **Historical Tracking:** Maintain records of past analyses for individual users.
- **Analytics:** Offer insights into crop health trends over time.

2.3: User Characteristics

User Characteristics: Users are expected to have basic computer literacy and access to either a smartphone or desktop computer.

2.4: Constraints

Constraints: The system must operate efficiently even under limited network conditions.

2.5: Assumptions/Dependencies

Assumptions/Dependencies: The system relies on a pre-existing database of labeled leaf images for training purposes.

3: Specific Requirements

3.1: External Interface Requirements

- **User Interface:** Mobile app interface and web portal for uploading images.
- **Hardware Interface:** Compatibility with various smartphones and computers.
- **Software Interface:** Integration with existing farm management systems if available.
- **Communication Interface:** Support for slow internet connections without compromising functionality.

3.2: Functional Requirements

FR-01 Image Upload

Input: Leaf images captured by farmers.

Processing: Store images temporarily until processed.

Output: Confirmation message upon successful upload.

Priority: Medium

FR-02 Image Analysis

Input: Uploaded leaf images.

Processing: Apply ML algorithms to classify images as healthy or diseased.

Output: Identification of disease(s) along with a confidence score.

Priority: High

FR-03 Treatment Suggestions

Input: Identified disease(s).

Processing: Retrieve relevant treatment advice from a knowledge base.

Output: List of suggested treatments or preventive measures.

Priority: High

FR-04 Historical Tracking

Input: User-specific data.

Processing: Record details of each image analysis session.

Output: Storage of historical records accessible to the user.

Priority: Low

FR-05 Analytics

Input: Historical data from user sessions.

Processing: Generate reports on crop health trends.

Output: Visualizations showing changes in crop health over time.

Priority: Medium

Non-Functional Requirements

Usability: The system shall be easy to navigate and understand for non-expert users.

Reliability: Ensure consistent accuracy in disease detection across different types of devices and networks.

Performance: Deliver results within a reasonable timeframe regardless of network speed.

Portability: Function well on both mobile and desktop platforms.

4: System Features

Feature 1: Image Upload

- Allows farmers to upload leaf images directly from their device's camera or photo gallery.
- Ensures compatibility with a wide range of smartphone and computer formats.

Feature 2: Image Analysis

- Utilizes advanced ML techniques to accurately diagnose plant diseases.
- Provides detailed explanations of how the diagnosis was made.

Feature 3: Treatment Suggestions

- Offers tailored solutions based on the specific disease detected.
- Includes links to external resources for further reading.

Feature 4: Historical Tracking

- Maintains a comprehensive log of all past analyses performed by the farmer.
- Enables users to review previous findings easily.

Feature 5: Analytics

- Generates visual summaries of crop health status over time.
- Helps farmers make informed decisions regarding future farming practices.

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